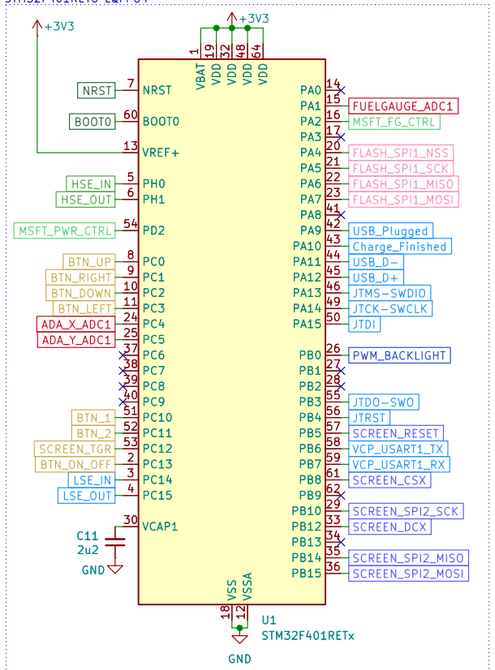
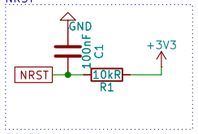
## MCU

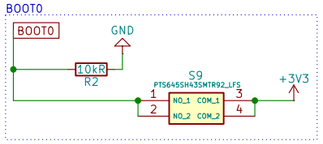
### STM32F401RET6 LQFP64



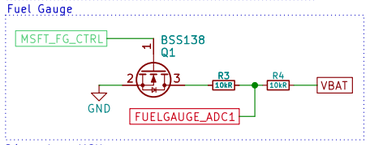

### NRST



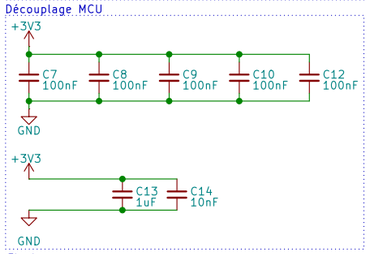

### BOOT0



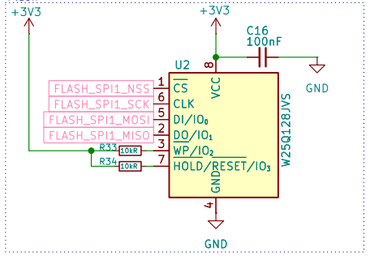

### Fuel Gauge




### Découplage MCU

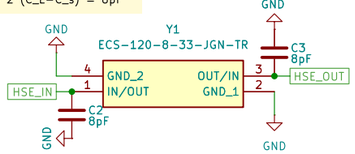



### Flash

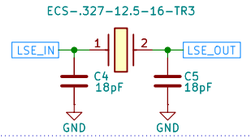



### Quartz Oscillator

Calcul des capacités HSE:
C_L = 8pF
C_s = 4pF
C = 2*(C_L-C_s) = 8pF

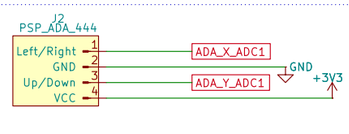



Calcul des capacités LSE:
C_L = 12.5pF
C_s = 4pF
C = 2*(C_L-C_s) = 17pF

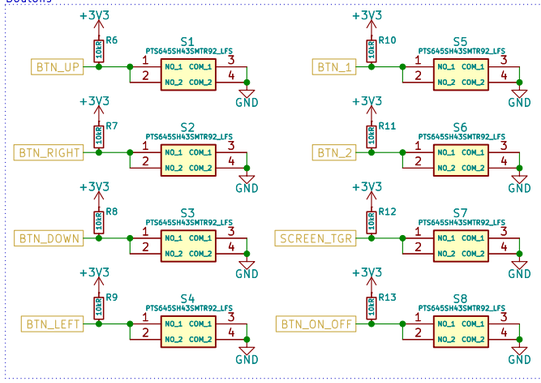



## Inputs

### PSP Joystick

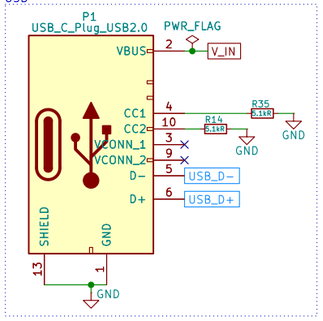



### Boutons

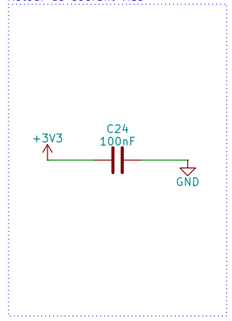



## USB

### USB

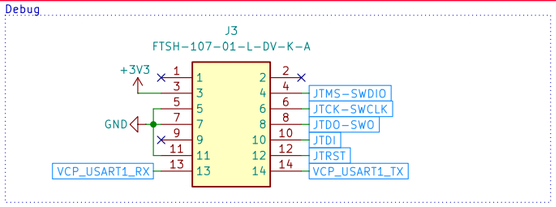



### Retour de courant vias

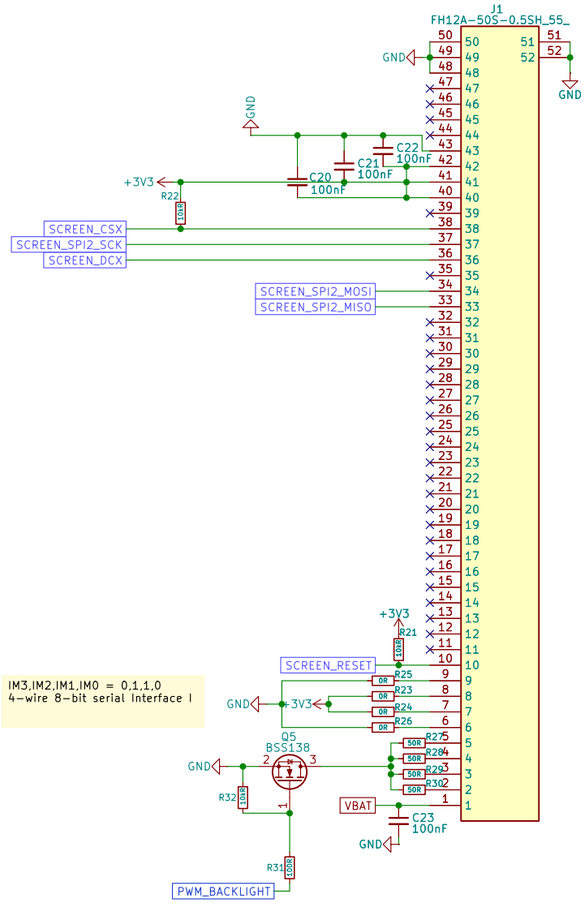



## Debug

### Debug

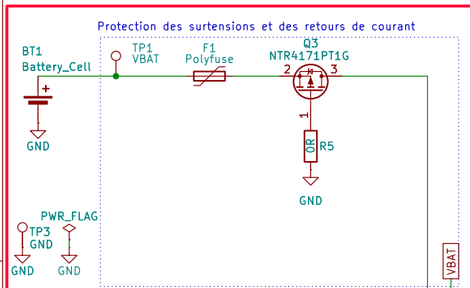



## Ecran

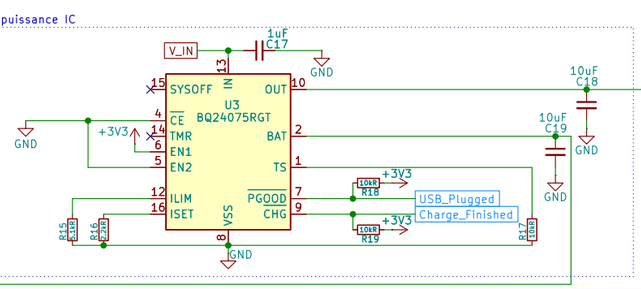



## ALIMENTATION

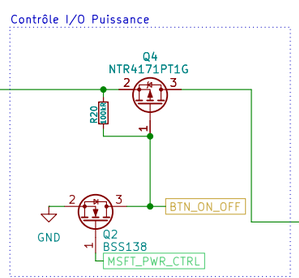
### Protection des surtensions et des retours de courant



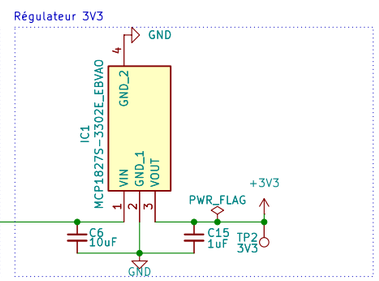

### Gestion de puissance IC




### Contrôle I/O Puissance




### Régulateur 3V3




| | |
|---|---|
| Sheet: / | |
| File: spatia-f4.kicad_sch | |
| **Author: Mathis Steinberger** | |
| Size: A3 | Date: 2026-08-08 |
| KiCad E.D.A. 10.0.5 | |